

PAPER_677: UQFF Predictions for the LISA Space-Based Gravitational Wave Observatory

Author: Daniel Murphy
Framework: Unified Quantum Field Framework (UQFF) v5.30
Session: 173
Date: April 1, 2026
Class: UQFFPredictionsForLISA | CP4 #261
Source: grok_share_fc21e30c24b4.txt
Domain: Space-Based GW / LISA

Abstract

The Laser Interferometer Space Antenna (LISA) will open the millihertz GW window, targeting SMBH mergers, EMRIs, and galactic binaries. The UQFF introduces a unique LISA-band suppression via the Aether arm-length modulation: $S_{\text{UA,LISA}} = 1 - \rho_{\text{UA}}L_{\text{LISA}}/(k_B T_{\text{eff}})$ and predicts an enhanced EMRI capture rate $R_{\text{EMRI,UQFF}} = R_{\text{GR}}(1 + f_{\text{TRZ}}\rho_{\text{UA}}/\rho_{\text{SCm}})$.

Primary UQFF Equation

$$h_{\text{UQFF,LISA}}(f) = h_{\text{GR}}(f) \cdot (1 - f_{\text{TRZ}}) \cdot S_{\text{UA,LISA}} \cdot S_{\text{SCm}}$$

Parameters

Parameter	Value	Description
L_{LISA}	2.5 Gm	LISA arm length
f	10^{-4} -1 Hz	LISA band
M_c	$10^8 M_{\odot}$	SMBH chirp mass

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m ³	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m ³	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor

Constant	Value	Description
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFPredictionsForLISA.h"
```

```
UQFFPredictionsForLISA obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFPredictionsForLISACalculator

calc = UQFFPredictionsForLISACalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

Amaro-Seoane et al. 2017 (LISA); UQFF Session 173

UQFF v5.30 | Session 173 | April 1, 2026 | PAPER_677 of 1000